RE

A language $L \subseteq \Sigma^*$ is called **regular** if $L = L(M)$ for some DFA M, where $L(M)$ is the language produced by that DFA M.

That is, a language is **regular** if every string in it can be accepted by one DFA.

Regular languages are closed under:

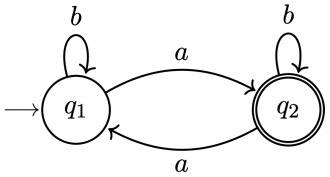
- Complement, if L is regular, then $\Sigma^* \backslash L$ is regular.
- Union, if L_1 and L_2 are regular, then $L_1 \cup L_2$ is regular.
- Intersection, if L_1 and L_2 are regular, then $L_1 \cap L_2$ is regular.
- Concatenation, if L_1 and L_2 are regular, then $L_1 L_2$ is regular.
- Kleene star, if L is regular, then L^* is regular.

For $\Sigma = \{a, b\}$, $(a|b)^*a$ means any string ending in a.

DFA

Deterministic finite automata, has:

- Q: finite set of states
- Σ: alphabet
- δ: $Q \times \Sigma \to Q$: transition function
  if $\delta(q, a) = q'$, we write $q \xrightarrow{a} q'$, a transition
- $q_0 \in Q$: initial state
- $F \subseteq Q$: final/accepting states



For example, this DFA accepts all strings that end in an a.

State	a	b
q_1	q_2	q_1
q_2	q_2	q_1

Note the transition function can be written as a table.

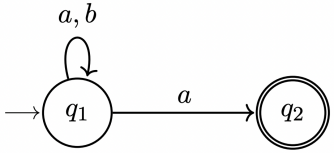
- Membership Problem: For a DFA M and a string w, decide if $w \in L(M)$.
- Non-emptiness Problem: For a DFA M, decide if $L(M) = \varnothing$.
- Equivalence Problem: For two DFAs M_1 and M_2, decide if $L(M_1) = L(M_2)$.

NFA

Nondeterministic finite automata, has the same ingredients as DFA. NFA is a DFA except states can have zero, one, or more outgoing transitions on the same input symbol.

We also allow epsilon-transitions, which are transitions that do not use the next symbol.

The input is accepted if at least one of it's runs is accepting.

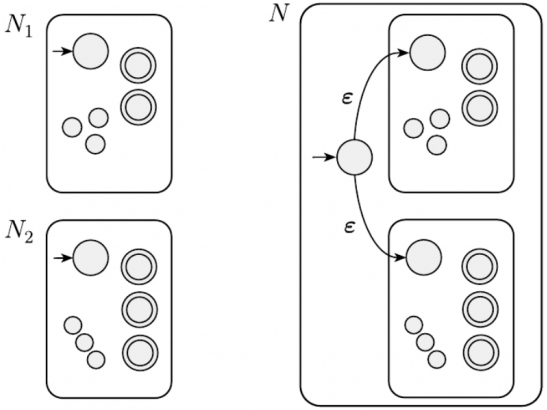


For example, this NFA accepts all strings that end in an a.

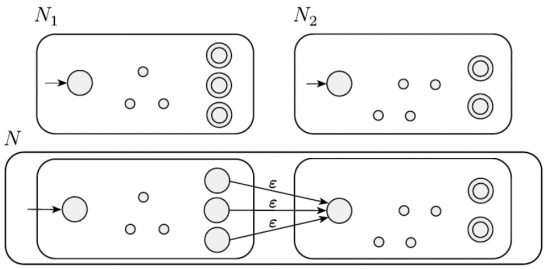
For every DFA M, there is at least one NFA N where $L(M) = L(N)$.

NFAs are closed under complement, union, intersection, concatenation and Kleene star. NFAs only recognise RE.

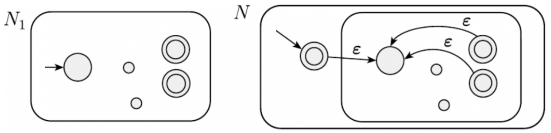
(Union) NFA N that recognises $L(N_1) \cup L(N_2)$:



(Concatenation) NFA N that recognises $L(N_1)L(N_2)$:

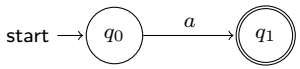


(Kleene star) NFA N that recognises $L(N_1)^*$:

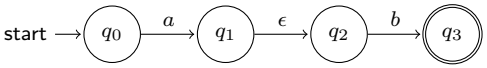


RE to NFA

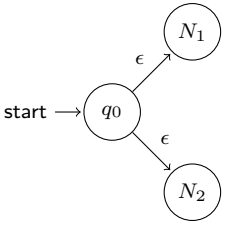
NFA for $R = a$, $a \in \Sigma$:



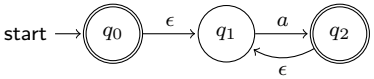
NFA for $R = ab$, $a, b \in \Sigma$ (concatenation):



NFA for $R = a \mid b$, $a, b \in \Sigma$ (union):



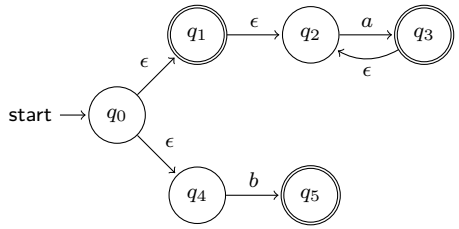
NFA for $R = a^*$, $a \in \Sigma$ (Kleene star):

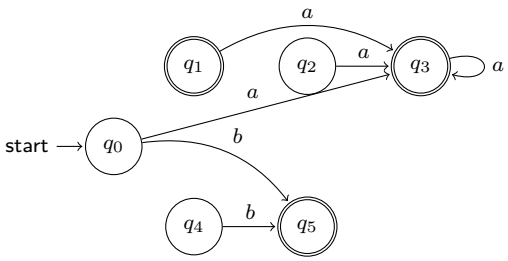


We can put these NFA's together to build an NFA that expresses the a required RE.

NFA to NFA without ϵ-transitions

Simply remove all epsilon-transitions, and add final states if necessary. For example:

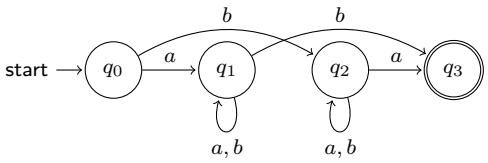




Note we do not remove states when removing epsilon transitions.

NFA without ϵ-transitions to DFA

We make the transition table of the NFA, then relabel states until we have a DFA. For example:



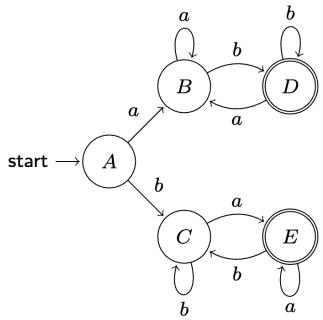
Original NFA:

State	a	b
q_0	q_1	q_2
q_1	q_1	$\boldsymbol{q_1, q_3}$
q_2	$\boldsymbol{q_2, q_3}$	q_2
$\boldsymbol{q_1, q_3}$	q_1	$\boldsymbol{q_1, q_3}$
$\boldsymbol{q_2, q_3}$	$\boldsymbol{q_2, q_3}$	q_2

Relabelling states to get DFA:

State	a	b
A	B	C
B	B	D
C	E	C
D	B	D
E	E	C

Start state is A, or q_0 from NFA. Accept states are D and E, since they had the accept state q_3 from NFA.

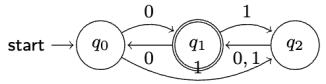


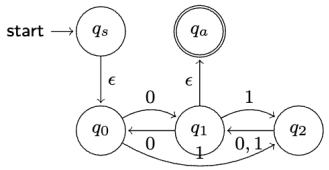
DFA to RE

First convert the DFA to a GNFA:

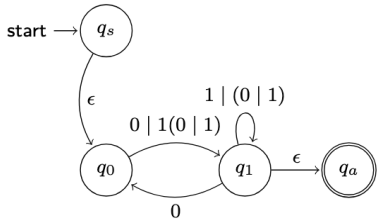
- Instead of a, b in a transition, write $a \mid b$.
- Add initial state and ϵ-transition to original initial state, and final states.

Then eliminate states until only 2 states left with 1 transition. For example:

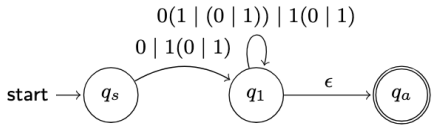




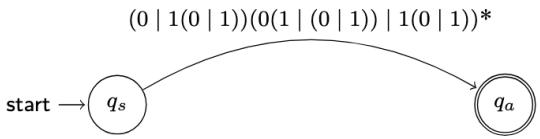
Eliminate state q_2:



Eliminate state q_0:



Eliminate state q_1:



So we get RE $(0 \mid 1(0 \mid 1))(0(1 \mid (0 \mid 1)) \mid 1(0 \mid 1))^*$

Prove a language is not regular

We can conclude $L(M) \neq L$ for a language L and any DFA M; if we can find strings x and y that go to the same state in M, then find another string z such that $xz \in L$ but $yz \notin L$.

That is there is no DFA for L and L is not regular.

For examample, $L = \{a^n b^n : n \geqslant 0\}$ is not regular:

1. Let $x_i = a^i$, $i \in \mathbb{N}$
2. For $i \neq j$, let $z = b^i$
3. Then $x_i z = a^i b^i \in L$
4. But $x_j z = a^j b^i \notin L$

CFG

Context free grammars, has:

- Variables, S, T
- Terminals (input symbols), a, b, c
- Rules, 3 in this case
- Start variable S

$S \to aSb$

$S \to T$

$T \to c$

The grammar derives strings of terminals as following:

1. Write the starting variable
2. Repeat the following until no variables are written:
   (a) Pick a variable X that is written down
   (b) Pick a rule $X \to ...$
   (c) Replace X with the RHS of the rule
3. The remaining string is derived from the grammar

For example: $S \implies T \implies c$

$S \implies aSb \implies aTb \implies acb$

Note a leftmost derivation always replaces the leftmost variable in the grammar at step 2(a).

Derivation notation:

$\implies$ derives in 1 step

$\overset{n}{\implies}$ derives in n steps

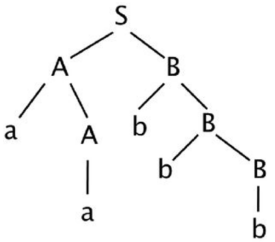
$\overset{0}{\implies}$ derives in 0 or more steps

$\overset{+}{\implies}$ derives in 1 or more steps

A language is called context-free if it can be derived from a CFG.

Parse Trees

- Root is labelled by start variable
- Nodes are labelled by variables
- Children are labelled by RHS of a rule
- Leaf nodes are a terminal or ϵ
- Read the string off the leaves, left to right



$S \to AB$
$A \to aA \mid a$
$B \to bB \mid b$
Parse tree for string $aabbb$

Ambiguous strings have $\geqslant 2$ parse trees in one CFG. A grammar is ambiguous if it generates at least 1 ambiguous string.

CFG to CNF

A CFG is in Chomsky Normal Form if every rule is in one of these forms:

1. $A \to BC$
2. $A \to a$
3. $S \to \epsilon$

We do **not** allow forms like $A \to 0A1$, $A \to B$, $A \to ab$, or the start variable on the RHS.

We can use any amount of | in a CNF.

Note parse trees of a grammar in CNF are binary trees.

Every CFG is guaranteed to have a CNF.

To turn a CFG into CNF:

1. START: Eliminate start variable from RHS of all rules
2. TERM: Eliminate rules with terminals (except those of form $A \to a$
3. BIN: Eliminate rules with more than 2 variables
4. EPSILON: Eliminate epsilon rules ($A \to \epsilon$), except $S \to \epsilon$
   If $B \to uAv$ and $A \overset{+}{\implies} \epsilon$, then $B \to uv$
5. Eliminate unit rules ($A \to B$)

CYK

Find if any CNF CFG accepts a string. Example, string $aabb$:

S, T			
$aabb$			
$\varnothing$	X		
aab	abb		
$\varnothing$	S, T	$\varnothing$	
aa	ab	bb	
A	A	B	B
a	a	b	b

For CNF CFG:
$S \to AB \mid AX \mid \epsilon$
$T \to AB \mid AX$
$X \to TB$
$A \to a$
$B \to b$

Runs in $O(|w|^3)$ time for fixed CFG, varying w.

TM

- Can only use variables (states) with finite domains
- Has infinite tape, made of cells
- There is a pointer/head on the tape
- The pointer can move left, right or stay still
- The pointer can read and write symbols from a bigger alphabet (tape alphabet, Γ)
- At first, the input string is written on the tape
- Depending on the state, the machine can decide to halt (stop running) and halt-accept or halt-reject
- Deterministic
- The language recognised by TM M is the set of strings it accepts

Each instruction in a TM is of form:

<**current state**><**current symbol**><**new symbol**><**direction**><**new state**>

E.g. **q1 a b R q2**, tells the TM if it reads an a and is in state q1, write a b, move right, and change to state q2.

Symbols are the tape alphabet Γ, which include the input alphabet Σ and blank _

Directions are L, R, and * (stay still, or S).

The machine stops when it reaches any state starting with "halt", i.e. halt-accept, halt-reject.

A TM can fail to accept an input for 2 reasons:

1. its computation is **rejecting**, i.e. halt-reject
2. its computation is **diverging**, i.e. it never halts

A language is **Turing-recognisable** if some TM M recognises it, that is it accepts all inputs in $L(M)$, but does not halt for all other inputs.

A language is **Turing-decidable** if some decider (TM that halts on all inputs) recognises it.

A TM runs in time $f(n)$ if $f(n)$ is the largest number of steps taken by the TM for any input of length n.

Must-move TM: Head cannot stay, must move. Every basic TM is equivalent to a must-move TM.

Left-bounded TM: Head starts at left-most cell of the tape, and the head cannot move further left than that point. Every basic TM is equivalent to a left-bounded TM.

Multitape TM: Multiple tapes, each with their own head, heads move simultaneously. Every basic TM is equivalent to a multitape TM.

Decidable languages are closed under complement, union and intersection.

A language is decidable when it and its compliment are recognisable.

NTMs: Nondeterministic TM, multiple possible transitions at one state, kinda like NFA.

P vs NP

P: Collection of languages decidable in polynomial time on **deterministic TMs**. Every RE and context-free language is in P. DFA membership, RE membership, DFA equivalence, CFG membership and CFG emptiness is in P.

NP: Collection of languages decidable in polynomial time on **nondeterministic TMs**. All languages in P are in NP. The CLIQUE problem is in NP. The **halting problem** (given an algorithm/program, find if it will ever halt) is in NP.

Propositional Logic

$\wedge$ conjunction (and), $\vee$ disjunction (or), $\neg$ negation (not). $\to$ implication, $\leftrightarrow$ bi-implication. $\top$ true. $\bot$ false.

An atom is a variable, and a formula is recursively by:

1. Every atom is a formula
2. If F is formula, $\neg F$ is formula
3. If F, G are formula, $F \vee G$ and $F \wedge G$ are formulas

F	G	$(F \to G)$
0	0	1
0	1	1
1	0	0
1	1	1

A formula is **valid** if every assignment satisfies it.
A formula is **satisfiable** if at least one assignment satisfies it.
Satisfiability problem, decide if a formula is satisfiable, is in NP.

To show F is a logical consequence $\models$ of $E_0, ..., E_k$ we use truth tables, see if every assignment that satisfies all formulas E_i also satisfies F.

NNF: Negation Normal Form, where $\neg$ negations only occur in front of atoms

CNF: Conjunctive Normal Form. Conjunction $\wedge$ of clauses (literal or disjunction $\vee$ of literals, a literal being p or $\neg p$).

Predicate Logic

Instead of "x is even" we write even(x). We cannot compose predicates, e.g. odd(prime(x)) is not a formula and has no meaning. We can use propositional logic syntax.

We have two quantifiers, $\exists x F$, existential qualifiers (there is an element d in the domain that F is true when d replaces x), and $\forall x F$ for all, universal qualifiers (for all elements d in the domain, F is true when d replaces x).

Bound and free variables ex: $\forall \underline{\boldsymbol{x}}(P(\boldsymbol{x}, y) \to \exists \underline{\boldsymbol{y}} Q(\boldsymbol{x}, \boldsymbol{y}, z))$